higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

COMMUNICATION-ELECTRONICS N6

9 APRIL 2019

This marking guideline consists of 12 pages.

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QUESTION 1

1.1

$$\text{Series L} = \frac{Z_o\,(f_2 - f_1)}{2.\pi.f_1.f_2}$$

$$= \frac{600(10000 - 7500)}{2 \times 3{,}142 \times 10 \times 10^3 \times 7{,}5 \times 10^3}$$

= 3,183 mH (1)

$$\text{Series C} = \frac{1}{2.\pi.Z_o\,(f_2 - f_1)}$$

$$= \frac{1}{2 \times 3{,}142 \times 600 \times (10000 - 7500)}$$

= 106,09 nF (1)

$$\text{Shunt L} = \frac{Z_o}{4.\pi.(f_2 - f_1)}$$

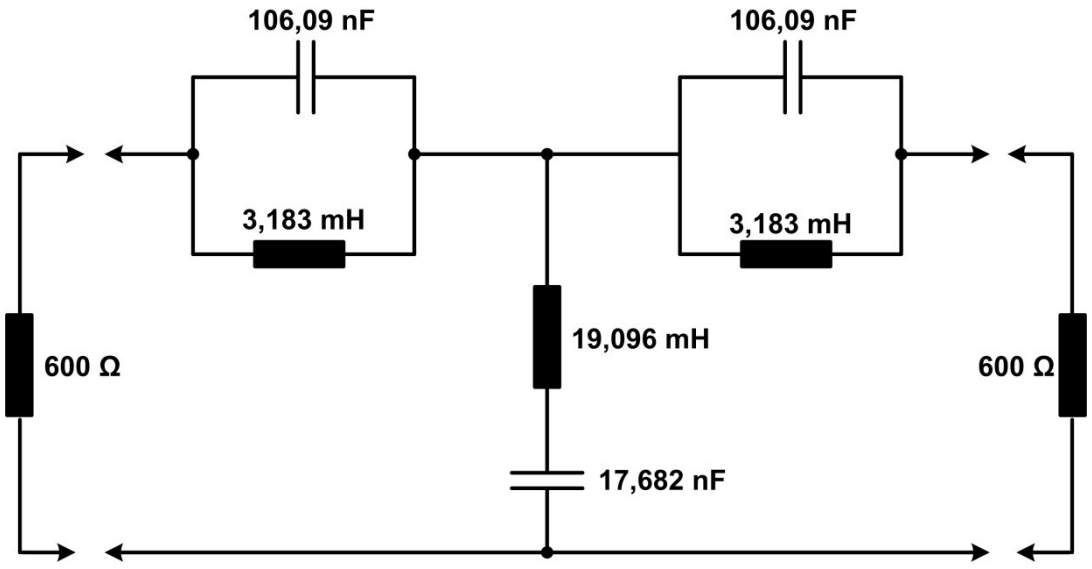
$$= \frac{600}{4 \times 3{,}142 \times (10000 - 7500)}$$

= 19,096 mH (1)

$$\text{Shunt C} = \frac{f_2 - f_1}{\pi.Z_o.f_1.f_2}$$

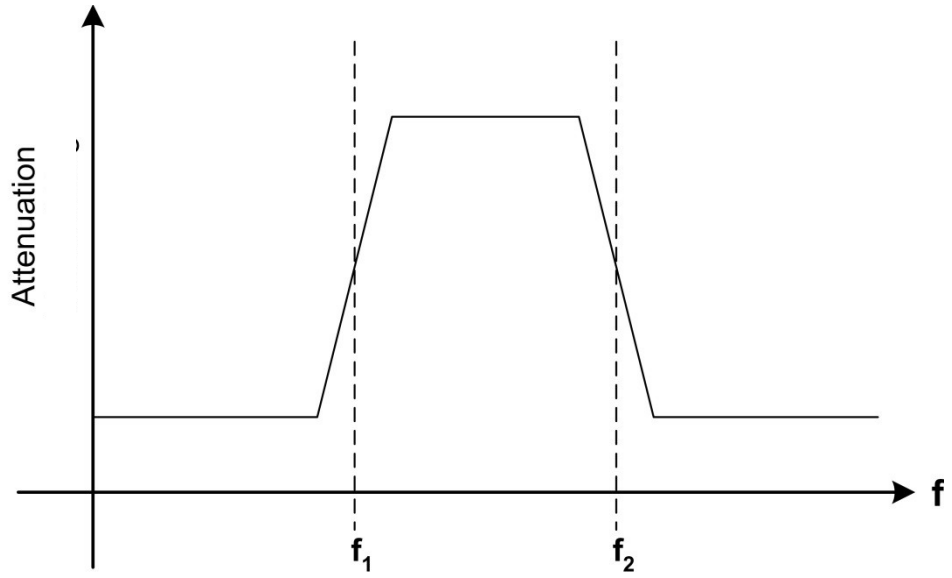
$$= \frac{10000 - 7500}{3{,}142 \times 600 \times 10000 \times 7500}$$

= 17,682 nF (1)

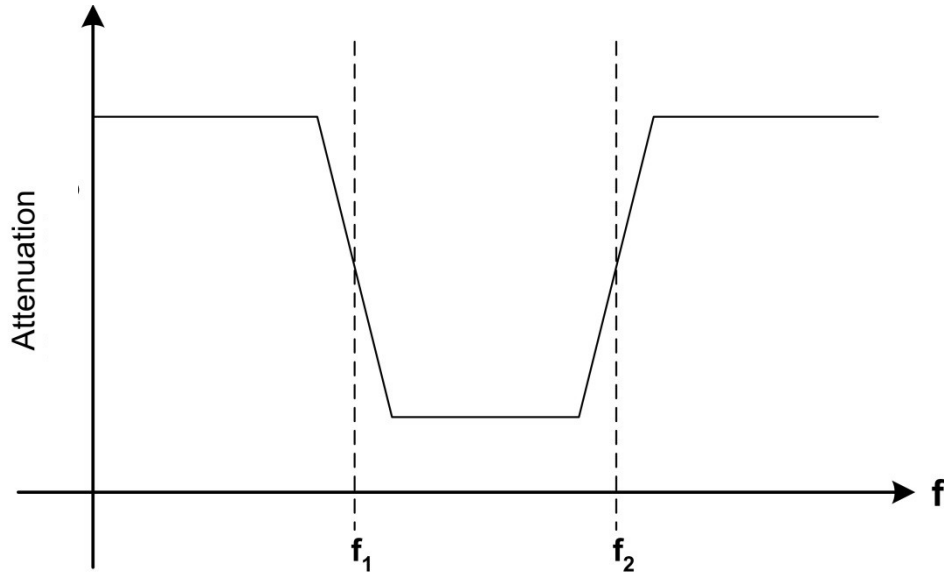


(2)

1.2 1.2.1



1.2.2



(2 × 2) (4)

1.3 $$L = \frac{Z_O}{4.\pi.fc}$$

$$= \frac{600}{4 \times 3{,}142 \times 3 \times 10^3}$$

=

= **15,913 mH** (1)

$$C = \frac{1}{4.\pi.Z_O.fc}$$

$$= \frac{1}{4 \times 3{,}142 \times 600 \times 3 \times 10^3}$$

=

(1)

= **44,204 nF**

Serie C = $\frac{C}{m}$

$= \frac{44,204 \times 10^{-9}}{0,6}$

= **73,673 nF** (1)

Shunt L = $\frac{2.L}{m}$

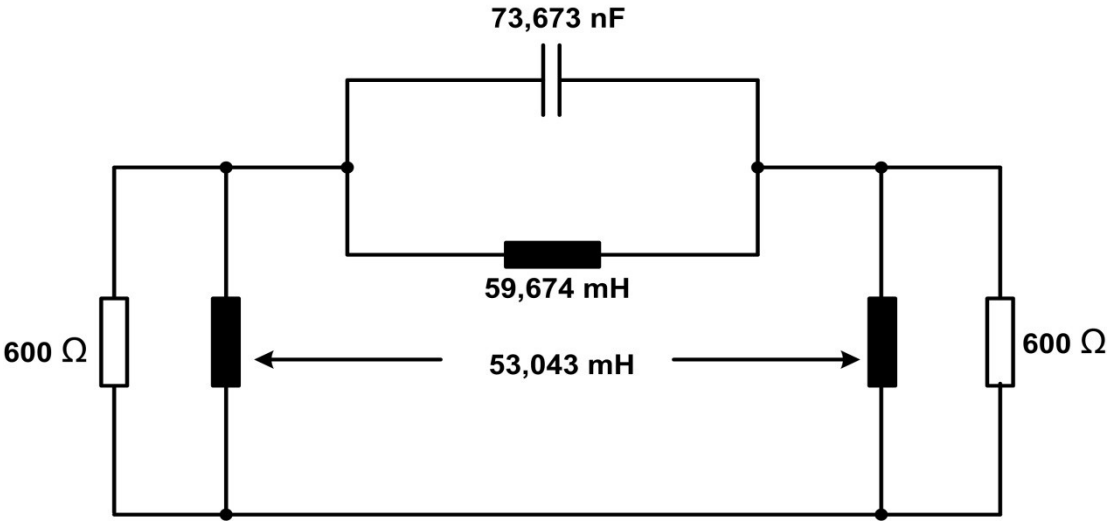
$= \frac{2 \times 15,913 \times 10^{-3}}{0,6}$

= **53,043 mH** (1)

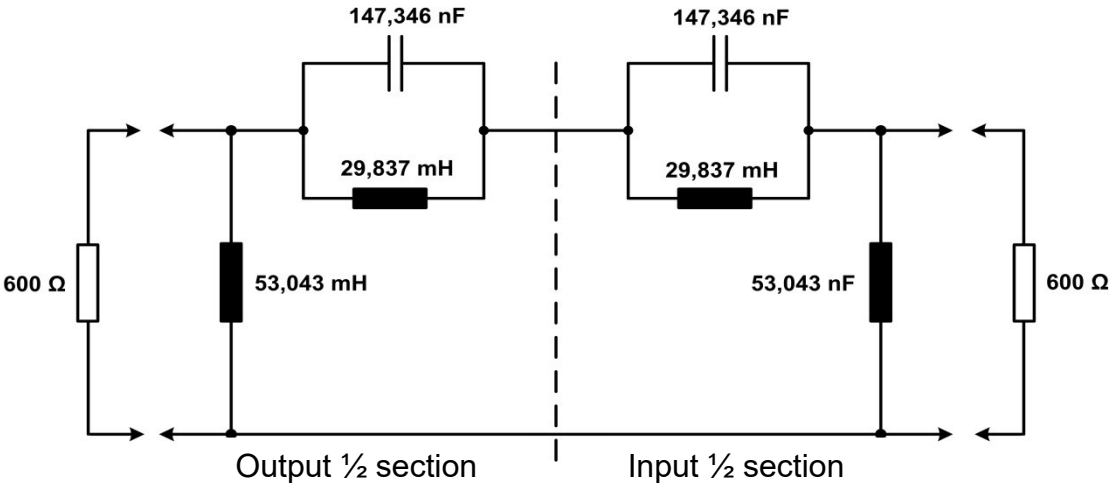
Serie L = $\frac{4.m}{1-m^2} \times L$

$= \frac{4 \times 0,6}{1-0,6^2} \times 15,913 \times 10^{-3}$

= **59,674 mH** (1)



(1)



(1)
[17]

QUESTION 2

2.1

$$N = \frac{I_{(in)}}{I(out/uit)}$$

$$= \frac{3}{1}$$

= 3 (1)

$$Z_1 = R_G\left[\frac{N^2+1}{N^2-1}\right] - 2\sqrt{R_G.R_L}\left[\frac{N}{N^2-1}\right]$$

$$= 560\left[\frac{3^2+1}{3^2-1}\right] - 2\sqrt{560\times 480}\left[\frac{3}{3^2-1}\right]$$

= 311,156 ohms (1)

$$Z_2 = 2\sqrt{R_G.R_L}\left[\frac{N}{N-1}\right]$$

$$= 2\sqrt{560\times 480}\left[\frac{3}{3^2-1}\right]$$

= 388,844 ohms (1)

$$Z_3 = R_L\left[\frac{N^2+1}{N^2-1}\right] - 2\sqrt{R_G.R_L}\left[\frac{N}{N^2-1}\right]$$

$$= 480\left[\frac{3^2+1}{3^2-1}\right] - 2\sqrt{560\times 480}\left[\frac{3}{3^2-1}\right]$$

= 211,156 ohms (2)

2.2

$$S = \sqrt{\frac{R_1}{R_2}}$$

$$= \sqrt{\frac{350}{500}}$$

= 0,837 (2)

$$1\left[\frac{N^2 - 1}{N^2 - 2.N.S + 1}\right]$$

$$= 350\left[\frac{15^2 - 1}{15^2 - (2 \times 15 \times 0{,}837) + 1}\right]$$

(2)

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=390,263 ohms

$$Z_B = \frac{\sqrt{R_1.R_2}}{2} \cdot \left[\frac{N^2 - 1}{N}\right]$$

$$= \frac{\sqrt{350 \times 500}}{2} \times \left[\frac{15^2 - 1}{15}\right]$$

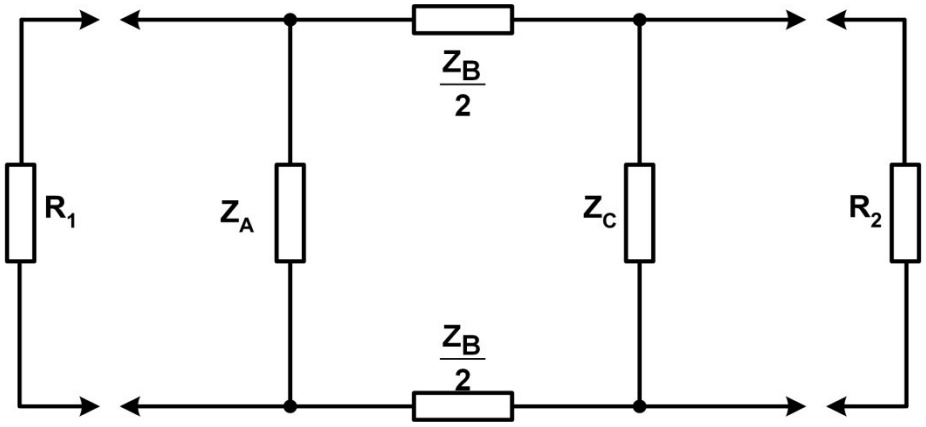
=3,124 k-ohms (2)

$$Z_C = R_2\left[\frac{N^2 - 1}{N^2 - 2.\frac{N}{S} + 1}\right]$$

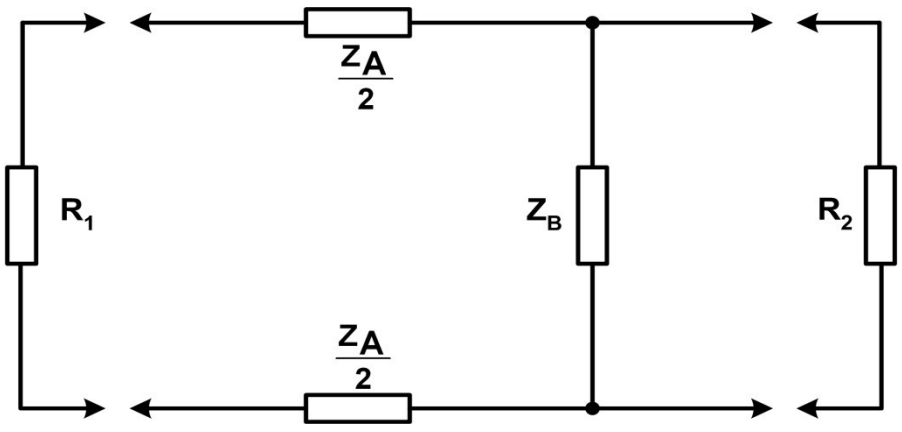
$$= 500\left[\frac{15^2 - 1}{15^2 - \left(2 \times \frac{15}{0,837}\right) + 1}\right]$$

= 588,985 ohms (2)

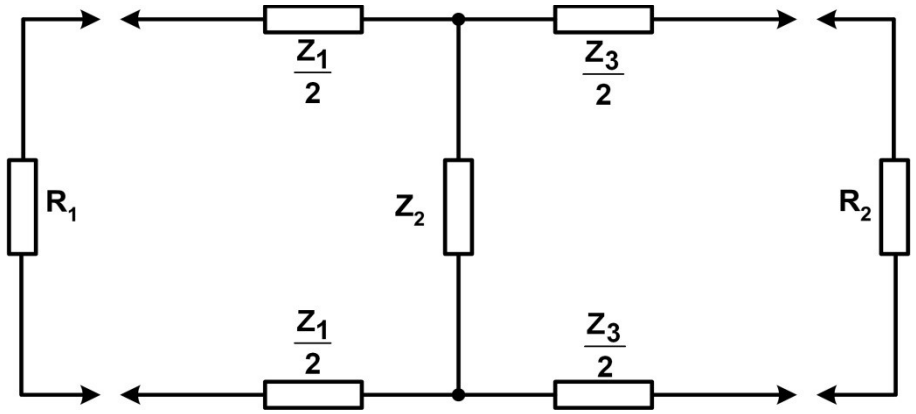
2.3 2.3.1



2.3.2



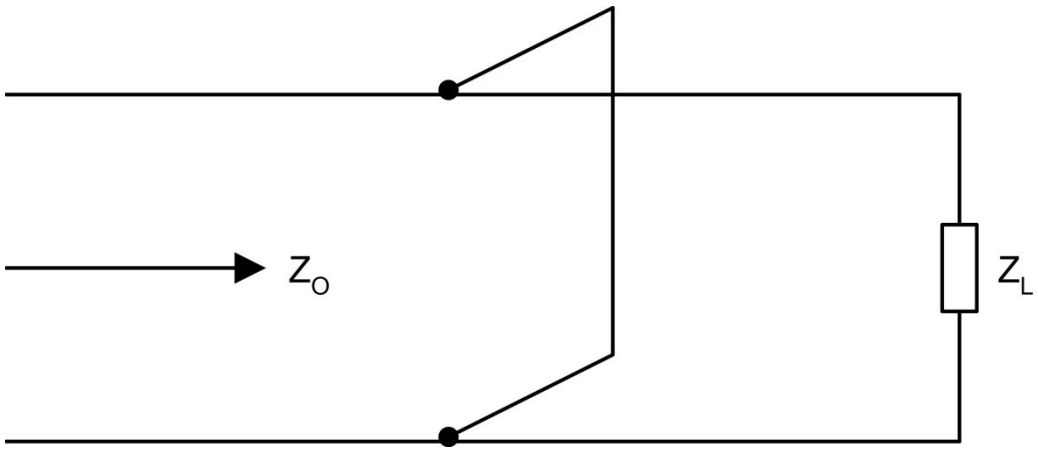
2.3.3



(3 × 1) (3)
[16]

QUESTION 3

3.1 Make use of a stub constructed as shown below:



The stub is a short piece of wire which is short-circuited across the transmission line. (4)

3.2
- Normalise impedance.
  $Z_O = \frac{R}{Z_o} \pm j\frac{X}{Z_o}$
- Plot R on the centre line and X on the circle.
- Follow the two lines until they intersect each other and plot this point.
- Draw the circle with a radius equal to the centre of the chart and the point plotted in the step above.
- Read off the SWR at the bottom of the circle where it intersects the centre line. (5)

3.3 3.3.1 α is the attenuation constant measured in nepers/km
3.3.2 β is the phase constant measured in radians/km

(2 × 2) (4)

3.4

$$Z_0 = \sqrt{Z_{o/c} . Z_{s/c}}$$

$$\therefore Z_{o/c} = \frac{Z_o^2}{Z_{s/c}}$$

$$= \frac{600^2}{750}$$

$$= \mathbf{480\ \Omega}$$ (3)

[16]

QUESTION 4

4.1 4.1.1

$$\mathbf{E_n} = \sqrt{4.K.T.\Delta F.(R_1 + R_2)}$$

$$= \sqrt{4 \times 1{,}38 \times 10^{-23} \times (273 + 25) \times 20 \times 10^3 \times (7{,}2 \times 10^3 + 5 \times 10^3)}$$

=2,003 μV (3)

4.1.2

$$\mathbf{R'_{eq}} = \left[R_{eq} - R_t\right] + \frac{R_{out/uit}}{A^2}$$

$$= \left[7{,}2 \times 10^3 - 5 \times 10^3\right] + \frac{18 \times 10^3}{20^2}$$

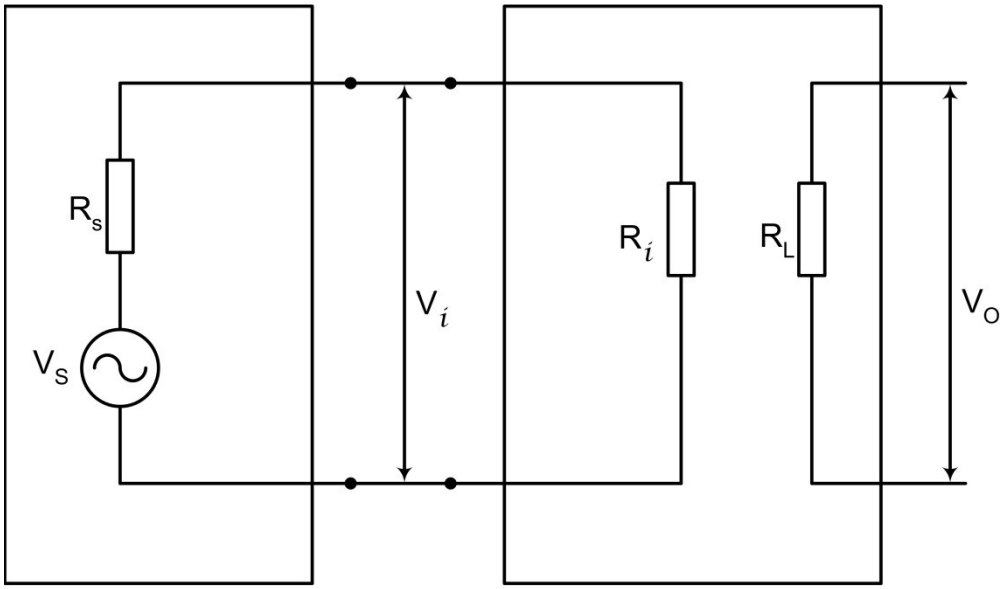
$$= 2{,}245\ k\Omega$$

$$\therefore F = 1 + \frac{R'_{eq}}{R_a}$$

$$= 1 + \frac{2245}{300}$$

= 8,483 (3)

4.2



GENERATOR **AMPLIFIER** (4)

4.3 Path loss = $2 \times 100 \times 0,002$

= 0,4 dB (1)

Path loss = $\text{antilog } \frac{\text{dB}}{10}$

$= \text{antilog } \frac{0,4}{10}$

= 1,096 (2)

$\lambda = \frac{c}{f}$

$= \frac{3 \times 10^8}{3 \times 10^9}$

= 0,1 m (2)

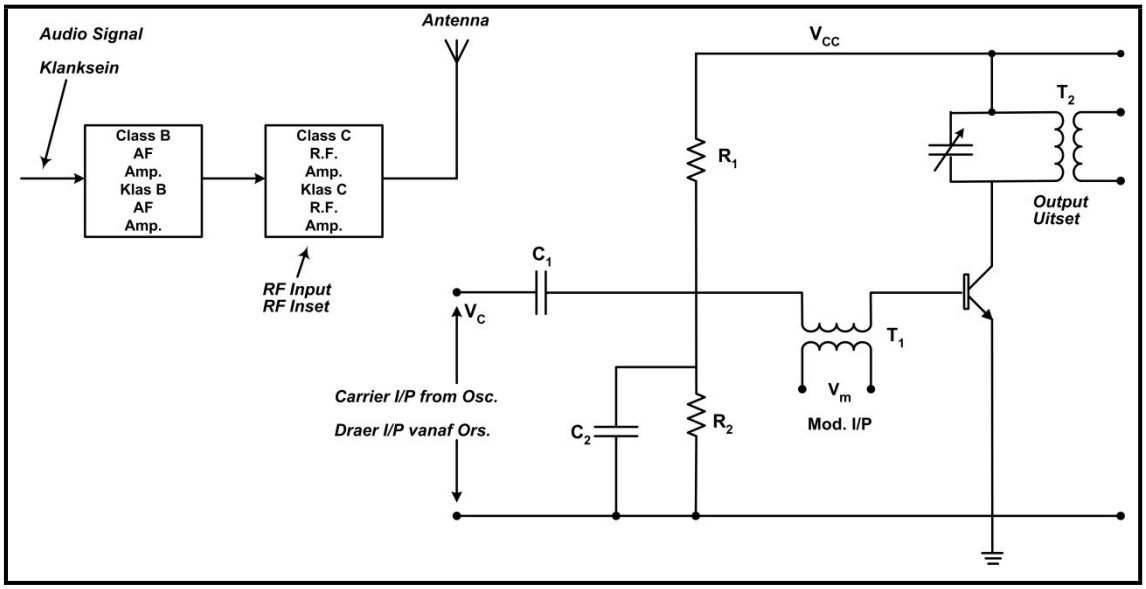
$P_t = \frac{[4\pi]^3 \cdot d^4 \cdot P_r}{G_t \cdot G_A \cdot Q \cdot \lambda^2} \times 1,096$

$= \frac{[4 \times 3,142]^3 \times [100 \times 10^3]^4 \times [1,8 \times 10^{-18}]}{20 \times 20 \times 9 \times 0,1^2} \times 1,096$

= 10,879 kW (2)
[17]

QUESTION 5

5.1



(7)

5.2 The carrier voltage V_C is applied to the base of Q_1 and is constant in the absence of a modulating signal V_m. Class C bias is supplied by the ratio R_1/R_2. When a modulating voltage V_m is applied via T_1 the base voltage will vary in sympathy and an AM output is obtained via T_2. (5)

5.3 5.3.1 The amplitude of the pulse is varied in proportion to the amplitude of the modulating signal.

5.3.2 The position of the pulse is varied in proportion to the amplitude of the modulating signal.

5.3.3 The width of the pulse is varied in proportion to the amplitude of the modulating signal.

(3 × 2) (6)

[18]

QUESTION 6

6.1 It is the angular separation between the two half-power points (-3 dB) on the radiation pattern of an antenna. (2)

6.2 6.2.1

$$\mathbf{E_S} = \frac{\sqrt{30.P_t.G}}{d}$$

$$= \frac{\sqrt{30 \times 60 \times 10^3 \times 15}}{150 \times 10^3}$$

$$= \mathbf{34,641 \times 10^{-3} volts/m^2} \quad (3)$$

6.2.2

$$P_d' = \frac{E_s^2}{120.\pi}$$

$$= \frac{\left(34,641 \times 10^{-3}\right)^2}{120 \times 3,142}$$

$$= \mathbf{3,183 \times 10^{-6} watts/m^2} \quad (2)$$

6.2.3

$$P_{reflected} = \frac{10.P_t.G}{4\pi d^2}$$

$$= \frac{10 \times 60 \times 10^3 \times 15}{4 \times 3,142 \times \left(150 \times 10^3\right)^2}$$

$$= \mathbf{31,827 \times 10^{-6} watts} \quad (2)$$

6.2.4

$$\mathbf{E_r} = \sqrt{\frac{30.P_{reflected}}{d}}$$

$$= \sqrt{\frac{30 \times 31,827 \times 10^{-6}}{150 \times 10^3}}$$

$$= \mathbf{79,783 \times 10^{-6} volts/m^2} \quad (2)$$

6.2.5

$$\mathbf{P_d} = \frac{E_r^2}{120.\pi}$$

$$= \frac{\left(79,783 \times 10^{-6}\right)^2}{120 \times 3,142}$$

$$= \mathbf{16,882 \times 10^{-12} watts/m^2} \quad (2)$$

6.2.6

$$\lambda = \frac{C}{f}$$

$$= \frac{3 \times 10^8}{950 \times 10^6}$$

$$= \mathbf{0{,}316\ m} \quad (1)$$

$$\mathbf{A} = \frac{\lambda^2 . G}{4.\pi}$$

$$= \frac{0{,}316 \times 15}{4 \times 3{,}142}$$

$$= \mathbf{0{,}377\ m^2} \quad (1)$$

$$\mathbf{P_r} = Pd\ .\ A$$

$$= 16{,}882 \times 10^{-12} \times\ 0{,}377$$

$$= \mathbf{6{,}365 \times 10^{-12}\ watts} \quad (1)$$

[16]

TOTAL: 100

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